What potential is there for bias or ethical issues when dealing with climate change data? Where would ClimateWins need to be cautious about using machine learning to develop answers? Consider the following questions and write an answer of around 200 words showing what pitfalls ClimateWins should avoid:

- **Is there personal information that may be exposed?**

In this kind of data that we are using, there is unlikely to be personal information or data that could be linked to individuals, since it will mostly be of observations of weather conditions.

Hoevere there is the possibility that each of those observations is attributed to an individual. IN that instance we would do one of two things:

1) Remove those details entirely when cleaning our data as they will not affect our findings
2) Alternatively, there may be a reason to believe that an individual may be important
   E.g. most observations are made my a small number of individuals using very different techniques.
   In this instance we would anonymise the individual themselves, but retain a flag to mark their identity. If we subsequently find that one person's results are drastically different to other people's, we may wish to exclude their results. This allows us to do that without needing to identify them.

- **Are there regional or cultural biases in climate change that might be made worse by using machine learning?**

It is probable that there is more data from more developed countries and regions, than from those that are less developed.

If all observations were treated equally, then there will be a skew towards the conditions in more developed countries – and certainly the northern hemisphere, that may not reflect the global south.

Culturally, we can expect at least some the data to come from the US, where we may still see imperial measures – we need to make sure these are converted to metric, or they will make our results inaccurate!

- **Is there human bias in climate change that might be propagated while training machine learning?**

Despite the overwhelming scientific evidence for climate change, there is still a great deal of political and social discussion about its effects, even if it exists.

We need to be careful with our data sources to ensure that we do not used biased data. Some data sets may have been very selectively chosen to further the views of the person using it.

- **Could machine learning potentially make incorrect decisions about where weather conditions might worsen and cause harm?**

Machine learning could well make incorrect decisions. The data we will be using is all historic, and climate change is happening rapidly. That change may start to affect weather patterns e.g. in the Caribbean and United States we are seeing the same amount of hurricanes, but a far greater likelihood of them being much more powerful. This has happened over a very few years. If we use tens or hundreds of years of data, this might appear a blip, rather than a trend/